EchoMaD-U-Net: Echocardiography Mask-conditioned Dehazing U-Net for Effective Echocardiography Denoising

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Abstract. Echocardiography is widely used in routine cardiology. However, noise frequently obscures fine structures making the images unusable for diagnosis. A repeat echocardiography examination may then be required. A key problem to solve, proposed by the Dehazing Echocardiography Challenge 2025, is to exploit all available data, both clean and noisy, to denoise images without sacrificing anatomical detail.
This paper presents *EchoMaD-U-Net*, a two-stage pipeline that combines robust anatomical segmentation with a lightweight U-Net denoiser. In the first stage, noisy two-dimensional transthoracic echocardiography images are segmented to generate region-of-interest (ROI) masks. State-of-the-art nnU-Net v2 is used for the first stage. In the second stage, the predicted masks are concatenated with the noisy images. Both are passed to a compact U-Net, which outputs a denoised image.
For the training of the second stage, a new dataset is created by adding noise to the clean images. A custom composite loss function is designed to balance global denoising quality, myocardial fidelity, and edge preservation.
Results demonstrate that the proposed *EchoMaD-U-Net* consistently improves chamber contrast and boundary definition even on very noisy scans, while maintaining computational efficiency. This underscores the potential of the model as a practical approach for robust echocardiography enhancement.

Keywords: Ultrasound dehazing · Echocardiography denoising · nnU-Net · U-Net · Segmentation

1 Introduction

Cardiac ultrasound is a widely used imaging modality to assess heart morphology and function due to its high availability, low cost and lack of ionising radiation.

However, transthoracic echocardiograms (TTEs) are commonly affected by noise which reduces image quality, visibility of fine structures and thus limits interpretability [1]. Naïve denoising methods often oversmooth clinically relevant details, whereas structure-aware methods require reliable localization of anatomical regions of interest (ROIs) [2]. Recently, nnU-Net v2 [3] has demonstrated strong performance across diverse cardiac imaging modalities [4,5] and enables reliable segmentation under noisy conditions with minimal manual tuning [6]. Building on this, we design *EchoMaD-U-Net*: Echocardiography Mask-Conditioned Dehazing U-Net, a two-stage approach in which segmentation informs denoising, allowing noise suppression without compromising myocardial boundaries.

Our contributions are summarized as follows:

1. A reproducible segmentation-conditioned denoising pipeline tailored for heterogeneous cardiac ultrasound data.
2. A structured training paradigm integrating noisy-label segmentation with simulated-noise denoiser training.
3. A composite loss function balancing global denoising quality, anatomical fidelity, and edge preservation.
4. A rigorous quality-control review of challenge training data to ensure reliability and consistency.

2 Method

2.1 Dataset and Preprocessing

The dataset is curated as part of the Dehazing Echocardiography Challenge 2025 to support the development of methods for dehazing and enhancing TTE images, with particular emphasis on patients who are more difficult to image. It consists of 4,500 four-chamber (4C) TTEs acquired from 75 patients who are considered relatively easy to image, and 2,400 4C TTEs acquired from 40 patients who are classified as difficult to image. Among the latter group, 237 images are provided with ROI segmentation masks of the ventricular septum and the left ventricular (LV) blood pool affected by speckle noise [1,7,8,9,10,11].

To ensure consistency and accuracy in downstream analysis, we perform a quality-control step by an interventional cardiologist (Dr. Pascal Lefebvre from CHU de Charleroi - ISPPC) to remove scans with erroneous segmentations. Specifically, 22 scans (9.2%) are excluded because the right ventricular (RV) blood pool is segmented instead of the LV blood pool, or the view corresponds to a five-chamber (5C) rather than the intended 4C orientation. Only erroneous segmentations are removed; no further grading of segmentation quality is performed.

After this exclusion step, 215 scans remain for model development and evaluation of the segmentation network. Table 1 summarizes the excluded scans and the reasons for their removal, and Figure 1 provides examples of erroneous cases, including an RV mis-segmentation and a 5C view, compared against a correctly labeled 4C scan that is retained for development.

Table 1. Excluded scans due to incorrect chamber segmentation (right ventricle segmented instead of the left ventricle) or inconsistent imaging view (5-chambers view vs. 4-chambers view).

Patient / Frame	Reason
patient 17 / frame 41	RV segmented
patient 20 / frame 1,21,41,51	RV segmented
patient 22 / frame 1,11,21,31,41,51	RV segmented
patient 24 / frame 1,11,21,31,41,51	5C view
patient 42 / frame 31	RV segmented
patient 44 / frame 1, 11	5C view
patient 48 / frame 11,21	RV segmented

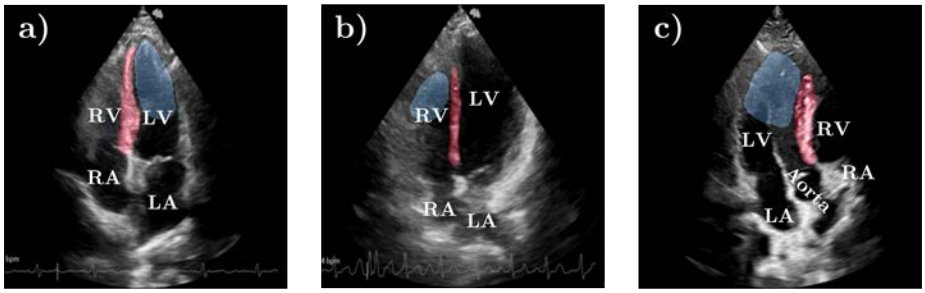

Fig. 1. Examples of excluded scans with erroneous segmentations compared to a retained image with good quality segmentations. Segmentation labels are shown in red for the ventricular septum and in blue for the LV blood pool region affected by speckle noise. (a) Correctly segmented 4C view (patient 13, frame 31). (b) Erroneous segmentation of the RV instead of the LV (patient 42, frame 31). (c) Misclassified 5C view (patient 24, frame 41).

The quality of the 4,500 four-chamber scans from the easy-to-image cohort is not independently assessed; instead, this subset is leveraged to construct an augmented dataset of synthetically noisy images. Noise is introduced using the `simulate_noise` function from [12], which applies StyleGAN-based, spatially specific perturbations. Three artifact types are modeled: global noise (layer 12, position 247), center-field noise (layer 9, position 416), and near-field noise (layer 12, position 100). The corresponding parameters, global, center-field and near-field are set to -2, 8, and 12, respectively, following an empirical parameter sweep to identify values that consistently generate realistic and visually plausible noise patterns across samples. These settings differ from those reported in the EchoNet Quality study [12]; they are selected to yield artifacts of comparable appearance to the provided noisy dataset and clinical relevance. Figure 2 illustrates representative outcomes across alternative parameter choices.

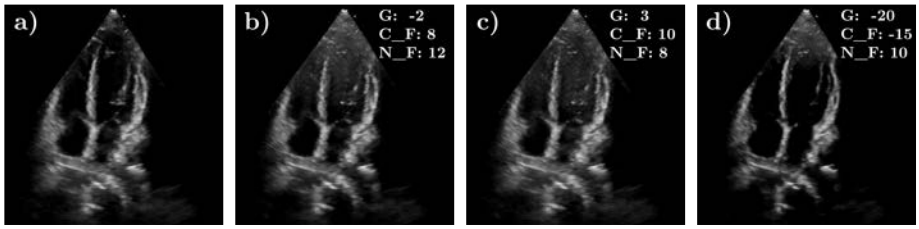

Fig. 2. Examples of synthetic noise generation on a clean echocardiography image using the `simulate_noise` function from [12]. (a) Original clean image (patient 1, frame 1). (b–d) Images with increasing levels of noise artifacts obtained by varying the global (G), center-field (C_F), and near-field (N_F) parameters.

2.2 Pipeline Overview

EchoMaD-U-Net consists of two sequential stages, illustrated in Figure 3, and is publicly available on GitHub at *EchoMaD-U-Net*:

1. **Segmentation:** Noisy 2D echocardiography frames are first processed with nnU-Net v2, employing cross-validated post-processing to generate reliable ROI masks.
2. **Mask-conditioned denoising:** The noisy image is concatenated with the predicted segmentation masks to form a two-channel input for a compact U-Net denoiser. It produces a single-channel denoised reconstruction while preserving anatomical fidelity.

2.3 Stage 1: Segmentation

Training Protocol. The first stage uses nnU-Net v2 and is trained on the 215 carefully triaged noisy images with provided ROI labels. Training is performed for 1000 epochs with 250 iterations per epoch, using a batch size of 10 across five-fold cross-validation. Validation consists of 50 iterations per epoch. The initial learning rate of 1×10^{-2} is adapted using a ReduceLROnPlateau scheduler, with weight decay set to 3×10^{-5}. Model selection relies on validation loss without early stopping, and Adam optimizer. Training is executed on a NVIDIA GeForce RTX 4070 Ti SUPER GPU.

Architecture. The network architecture for Stage 1 is depicted in Figure 3. The 2D PlainConvUNet accepts single-channel 256×256 images and comprises seven resolution levels with six downsampling operations. The network starts with 32 base features, doubling at each stage up to a maximum of 512 (32, 64, 128, 256, 512, 512, 512). Each encoder and decoder stage employs two consecutive 3×3 convolutions, with 2×2 max pooling. Standard U-Net skip connections link encoder and decoder stages, and deep supervision provides auxiliary outputs at multiple decoder scales. Batch normalization follows each convolution.

Loss Function. The training loss combined Dice loss and cross-entropy, computed over the entire batch.

Augmentation. Spatial augmentations include random rotations, scaling and elastic deformations. Intensity augmentations comprise Gaussian noise (variance ≤ 0.1), Gaussian blur ($\sigma = 0.5$–1.0), brightness and contrast adjustments (0.75–1.25), and gamma transformations (0.7–1.5). Low-resolution simulation is applied with probability 0.25 (scaling 0.5–1.0). Horizontal and vertical flipping, and foreground oversampling is performed with 33% probability to ensure adequate tissue representation.

2.4 Stage 2: Mask-conditioned denoising

Training Protocol. The U-Net denoiser is trained on 4,500 clean images degraded using the `simulate_noise` method from EchoNet-Quality [12]. Predicted segmentation masks from the pretrained nnU-Net v2 are provided as an additional input channel, allowing the denoiser to learn mappings from noisy image–mask pairs to the corresponding clean images. Optimization is performed with Adam, using a learning rate of 1×10^{-4}. Training is conducted on a NVIDIA GeForce RTX 4070 Ti SUPER GPU.

Architecture. The network architecture for Stage 2 is depicted in Figure 3. The denoiser employs a U-Net architecture with two input channels (noisy frame and mask) and a single output channel. The encoder starts with a double convolution from 2 to 32 channels, followed by four downsampling stages that doubles feature channels up to 512. The decoder mirrors the encoder, using bilinear upsampling, skip connections, and progressive reductions back to 32 channels. Each double convolution block comprises two consecutive 3×3 convolutions without bias, followed by batch normalization and ReLU activation. A final 1×1 convolution produces the denoised output.

Loss Function. Training uses a composite weighted loss designed to align with the challenge evaluation metrics, including Fréchet Inception Distance (FID), Contrast-to-Noise Ratio (CNR), generalized CNR (gCNR), Kolmogorov–Smirnov (KS) statistics on annotated regions, and Dice and Average Surface Distance (ASD) from downstream segmentation. The total loss is defined as

$$L_{\text{total}} = 5.0 \cdot L_{\text{denoise}} + 2.0 \cdot L_{\text{structure}} + 2.0 \cdot L_{\text{downstream}},$$

where L_{denoise} promotes global fidelity and perceptual realism (consistent with FID), $L_{\text{structure}}$ enhances myocardial preservation and contrast between ventricular regions (targeting CNR, gCNR, and KS), and $L_{\text{downstream}}$ sharpens edges to improve Dice and ASD in subsequent segmentation. Full mathematical definitions and weighting strategies are provided in the Supplementary Material 5.

Augmentation. Light geometric augmentations are applied, including random horizontal flips (p=0.5), vertical flips (p=0.1), and rotations within ±10°, to improve robustness while preserving anatomical structures.

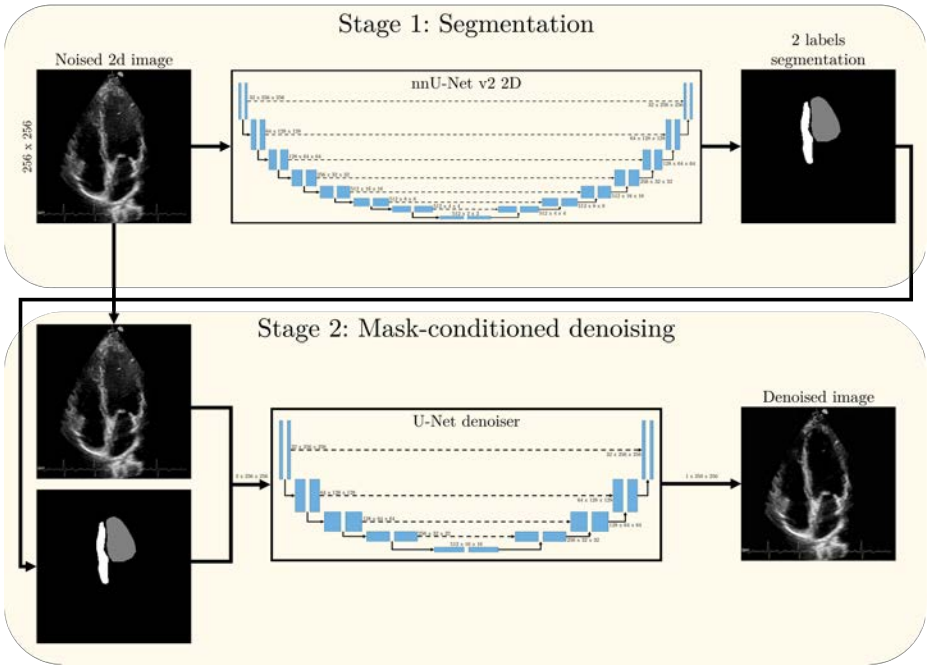

Fig. 3. *EchoMaD-U-Net* architecture. **Stage 1:** Noisy 4C echocardiography images are processed by nnU-Net v2 (2D configuration) to generate segmentation masks. The white masks delineate myocardial tissues of the ventricular septum, which should be preserved. The dark grey region corresponds to noise artifacts within the LV, where multipath reflection noise should be removed. **Stage 2:** The noisy image and its predicted segmentation mask are concatenated to form a two-channel input for a compact U-Net denoiser. It reconstructs a cleaned echocardiography image. The mask guides denoising to preserve myocardial boundaries and structural fidelity while effectively reducing speckle noise.

3 Results

3.1 Challenge Evaluation

The final submission of *EchoMaD-U-Net* demonstrates competitive performance across all official challenge metrics. Table 2 summarizes the aggregated scores obtained from the online evaluation platform. The model achieves a mean Dice score of 0.906 with low variance, indicating reliable segmentation consistency.

Although the FID is relatively high at 108.5, structural preservation remained strong, as reflected by high gCNR (0.841) and CNR (1.95) values. Surface accuracy is robust, with an average surface distance (ASD) of 1.80 mm. Statistical similarity between intensity distributions in annotated regions is further supported by KS scores. The overall challenge score of 75.0 is computed using a weighted combination of the four evaluation categories. Denoising performance (FID, CNR and gCNR), structural preservation (KS), and downstream task performance (Dice and ASD) are weighted in a 5:3:2 ratio, respectively.

Table 2. Aggregated evaluation metrics for *EchoMaD-U-Net* from the online evaluation platform (transposed). Lower is better for FID, ASD, and KS ROI1; higher is better for Dice, gCNR, CNR, KS ROI2 and Final Score.

	FID	Dice	gCNR	CNR	ASD	KS ROI1	KS ROI2	Final Score
Mean	108.5308	0.9058	0.8410	1.9514	1.7950	0.1431	0.7613	75.0332
Std	–	0.0524	0.0722	0.4777	1.1795	0.0584	0.0853	–

3.2 Best- and Worst-Case Performance

Figure 4 allows a qualitative analysis of the best and worst predictions according to the gCNR. In the best case (patient 14, frame 6), the denoised output preserves myocardial boundaries and ventricular contrast, achieving gCNR of 0.975 and CNR of 3.77. The worst case (patient 36, frame 57) exhibits residual noise and reduced contrast, with gCNR dropping to 0.577 and CNR to 1.03. This degradation correlates with poorer segmentation quality, underscoring the dependence of denoising performance on segmentation accuracy. Nevertheless, anatomical structures remain identifiable, illustrating the robustness of mask conditioning even in challenging cases.

3.3 Comparison with EchoNet-Quality

To assess the impact of segmentation conditioning, we compare *EchoMaD-U-Net* with the publicly available EchoNet-Quality denoising baseline [12] across 536 scans from the final test phase. Predicted masks are generated using segmentation Stage 1 of our proposed model, and evaluation is performed using the official challenge metrics. As ground truth labels are not provided for this set, Dice and ASD scores cannot be calculated.

As shown in Table 3, the proposed *EchoMaD-U-Net* outperforms EchoNet-Quality across most structural and statistical metrics. In particular, gCNR improves significantly (0.907 vs. 0.763), highlighting enhanced myocardial-to-blood pool contrast. KS statistics also demonstrates superior distribution alignment in annotated regions. Although EchoNet-Quality achieves a lower FID (80.4 vs. 119.2), suggesting slightly more naturalistic global image texture, our proposed method achieves a higher overall final score (77.1 vs. 71.9).

Figure 5 provides a qualitative comparison, highlighting the improvement in chamber contrast and boundary sharpness achieved by *EchoMaD-U-Net* relative to EchoNet-Quality. The arrows indicate regions of residual speckle noise that are better identified and suppressed by *EchoMaD-U-Net*, demonstrating the benefit of mask-conditioning in targeting noise within the ventricular cavity. The circles highlight sharper delineation of myocardial boundaries, where EchoNet-Quality tended to oversmooth and blur fine anatomical structures. These differences are most pronounced in cases with substantial speckle noise, where mask-conditioning effectively guided the network towards preserving anatomical fidelity.

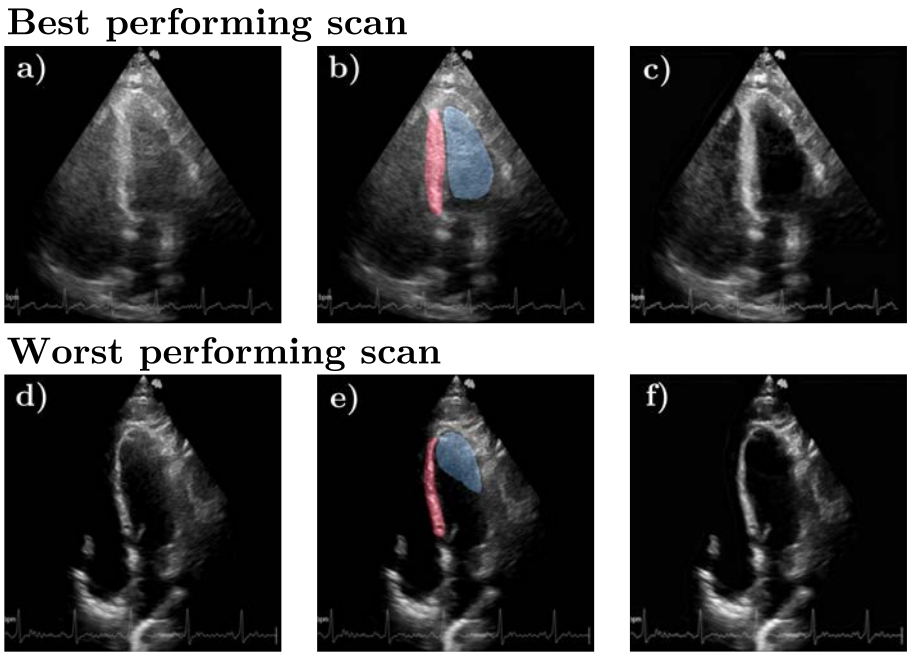

Fig. 4. Representative examples of *EchoMaD-U-Net* performance ranked by gCNR. **Best case**: patient 14, frame 6, shows clear myocardial boundaries and strong chamber contrast. **Worst case**: patient 36, frame 57, shows residual noise and lower contrast reduce structural fidelity. Panels: **a**) original noisy image (best case), **b**) predicted segmentations (best case), **c**) corresponding denoised output (best case), **d**) original noisy image (worst case), **e**) predicted segmentations (worst case), and **f**) corresponding denoised output (worst case).

Table 3. Comparison of evaluation metrics between *EchoMaD-U-Net* and Echonet on 536 test scans. Lower is better for FID and KS ROI1; higher is better for all other metrics.

Metric	**EchoMaD-U-Net**	**Echonet**
FID	119.2187	80.4029
CNR (mean ± std)	2.2271 ± 0.3815	1.5483 ± 0.3020
gCNR (mean ± std)	0.9065 ± 0.0413	0.7625 ± 0.0875
KS ROI1 (mean ± std)	0.1386 ± 0.0518	0.1793 ± 0.0709
KS ROI2 (mean ± std)	0.8265 ± 0.0711	0.6842 ± 0.1082
Final Score	77.1081	71.9253

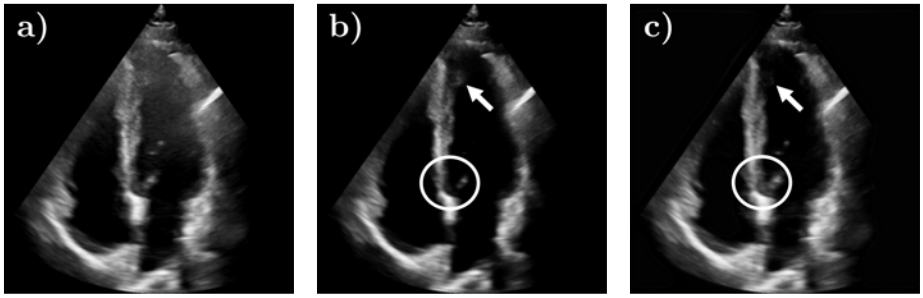

Fig. 5. Qualitative comparison of denoised outputs for a noisy test image (patient 37, frame 37). (**a**) Input noisy frame. (**b**) Denoised output from EchoNet-Quality, with smoother but less structurally faithful results. (**c**) Denoised output from *EchoMaD-U-Net*, showing sharper myocardial boundaries and improved chamber contrast. Arrows indicate regions of residual noise better suppressed by *EchoMaD-U-Net*, while circles highlight clearer delineation of myocardial boundaries.

4 Discussion

Our study introduces *EchoMaD-U-Net*, a two-stage mask-conditioned denoising framework designed for 4C TTE images. The method achieves competitive performance across metrics, demonstrating that segmentation-guided denoising can effectively preserve myocardial boundaries while improving chamber contrast. Importantly, the approach offers a lighter-weight alternative to generative models, such as EchoNet-Quality [12], relying on a compact U-Net architecture rather than generative adversarial training. This simplicity makes the framework particularly attractive for deployment in resource-constrained clinical settings.

One consistent observation is the trade-off between perceptual realism (generative adversarial methods) and structural fidelity (U-Net methods). While generative methods typically excel at reducing FID by generating textures that closely mimic natural image statistics, they often do so at the expense of anatomical precision. By contrast, *EchoMaD-U-Net* prioritizes accurate structural reconstruction, yielding smoother outputs that achieve high Dice, gCNR, and ASD performance despite higher FID values. For echocardiography, the diagnostic

accuracy depends on reliable delineation of anatomical structures rather than photorealistic textures, the proposed approach is preferable.

The design of the pipeline, i.e. segmentation first, denoising second, proves critical in stabilizing predictions. Conditioning denoising on predicted masks guides the network to preserve key boundaries, mitigating the tendency of purely image-based models to oversmooth regions of diagnostic interest. The use of simulated noise during training further enables robust generalization across diverse corruption scenarios, though real-world noise distributions may deviate from these simulations.

Clinical feedback further supports this interpretation. According to the practicing cardiologist Dr. Pascal Lefebvre, the images produced by *EchoMaD-U-Net* sometimes appear smoother, but remain acceptable for diagnostic use provided that essential features such as septal thickness are preserved. More importantly, the denoised outputs are reported to facilitate clearer identification of localized contractility abnormalities, potentially improving the detection of pathologies such as myocardial infarction. This confirms that anatomical fidelity, rather than strict perceptual realism, is the most relevant quality criterion for echocardiographic denoising.

Several limitations should be acknowledged. The current implementation reduces volumetric data to individual 2D frames, which may limit the ability to exploit temporal and spatial coherence inherent to echocardiographic sequences. Conditioning is restricted to two ROI masks rather than multi-class anatomical priors, potentially underutilizing structural information. Single-scale U-Net backbone is employed; multi-scale or diffusion-based priors could enhance restoration of fine-grained detail.

Beyond these technical considerations, there are practical implications. The proposed method achieves strong denoising performance with a simpler and computationally lighter backbone than EchoNet-Quality, although we recognize the usefulness of EchoNet-Quality to generate the noisy images. The efficiency of the proposed method makes it better suited for integration into clinical workflows or portable ultrasound devices, where hardware constraints and inference speed are important. Future work could explore replacing the nnU-Net segmentation stage with lighter-weight networks to further reduce complexity, or extending mask-conditioning into 3D and temporal architectures to leverage volumetric context.

5 Conclusion

In this paper, *EchoMaD-U-Net* is presented, a two-stage mask-conditioned denoising framework that combines nnU-Net v2 segmentation with a lightweight U-Net. The method provides effective denoising while preserving myocardial structures, offering a straightforward alternative to more complex generative adversarial approaches. Early feedback from a cardiologist indicates that key anatomical features and functional patterns remain visible with *EchoMaD-U-Net*, supporting practical use in echocardiography assessment. These results sug-

gest that segmentation-guided denoising is a useful and implementable approach for improving ultrasound image quality in research and clinical settings.

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Supplementary Material

S1. Detailed Loss Function Definitions

The training procedure for the UNet denoiser utilizes a composite weighted loss function designed to balance global denoising quality, anatomical fidelity in the myocardium, and edge/structure preservation. The loss is implemented with the following structure:

$$L_{\text{total}} = w_{\text{denoise}} \cdot L_{\text{denoise}} + w_{\text{structure}} \cdot L_{\text{structure}} + w_{\text{downstream}} \cdot L_{\text{downstream}} \quad (1)$$

The default group weights are set as:

$$w_{\text{denoise}} = 5.0, \quad w_{\text{structure}} = 2.0, \quad w_{\text{downstream}} = 2.0 \quad (2)$$

Loss Component Definitions

Global Denoising and Stability ($L_{denoise}$). This component enforces overall fidelity, perceptual realism, and statistical consistency:

$$L_{\text{mix}} = \alpha \cdot \text{L1}(\hat{x}, x) + (1-\alpha) \cdot [-\text{PSNR}(\hat{x}, x)], \quad \alpha = 0.8 \quad (3)$$

$$L_{\text{feat}} = \text{MSE}\big(\phi(\hat{x}_s), \phi(x_s)\big) \quad (4)$$

$$L_{\text{mean}} = \text{MSE}\big(\mu(\hat{x}), \mu(x)\big), \quad L_{\text{var}} = \text{MSE}\big(\text{Var}(\hat{x}), \text{Var}(x)\big) \quad (5)$$

$$L_{\text{denoise}} = L_{\text{mix}} + 0.05 \cdot L_{\text{feat}} + 0.02 \cdot L_{\text{mean}} + 0.05 \cdot L_{\text{var}} \quad (6)$$

where $\hat{x}$ denotes the prediction, x the clean target, $\phi(\cdot)$ is a lightweight feature extractor for perceptual similarity, and $\hat{x}_s, x_s$ are jointly min-max normalized versions scaled to $[0, 1]$.

Structure and Anatomy Preservation ($L_{structure}$). This component emphasizes myocardial preservation and region-specific contrast:

$$\text{L1}_{\text{ROI-A}} = \text{L1}(\hat{x}, x) \quad \text{within ROI-A (myocardium)} \quad (7)$$

$$\text{CNR} = \frac{|\mu_{\text{ROI-A}} - \mu_{\text{ROI-B}}|}{\sqrt{\sigma^2_{\text{ROI-A}} + \sigma^2_{\text{ROI-B}}}} \quad (8)$$

$$\text{B}_{\text{change}} = \text{L1}(\hat{x}, x_{\text{noisy}}) \quad \text{within ROI-B} \quad (9)$$

$$L_{\text{structure}} = 0.4 \cdot \text{L1}_{\text{ROI-A}} - 0.02 \cdot \text{CNR} - 0.0 \cdot \text{B}_{\text{change}} \quad (10)$$

where ROI-A corresponds to myocardium (mask label = 1, or any label > 0 for binary masks), ROI-B corresponds to region B (mask label = 2), and x_{noisy} is the original noisy input.

Downstream Task Optimization ($L_{downstream}$). This term sharpens edges to improve subsequent segmentation performance:

$$L_{\text{grad}} = \text{L1}\big(\nabla\hat{x}, \nabla x\big) \quad \text{within ROI-A} \quad (11)$$

$$L_{\text{downstream}} = 0.02 \cdot L_{\text{grad}} \quad (12)$$

where ∇ denotes Sobel gradient computation.

Complete Formulation Combining all components with their respective weights:

$$\begin{aligned} L_{\text{total}} = 5.0 \cdot \Big[& \alpha \cdot \text{L1} + (1-\alpha)(-\text{PSNR}) + 0.05 \cdot L_{\text{feat}} + 0.02 \cdot L_{\text{mean}} + 0.05 \cdot L_{\text{var}} \Big] \\ & + 2.0 \cdot \Big[0.4 \cdot \text{L1}_{\text{ROI-A}} - 0.02 \cdot \text{CNR} - 0.0 \cdot \text{B}_{\text{change}} \Big] \\ & + 2.0 \cdot \Big[0.02 \cdot L_{\text{grad}} \Big], \quad \text{with } \alpha = 0.8 \end{aligned} \tag{13}$$